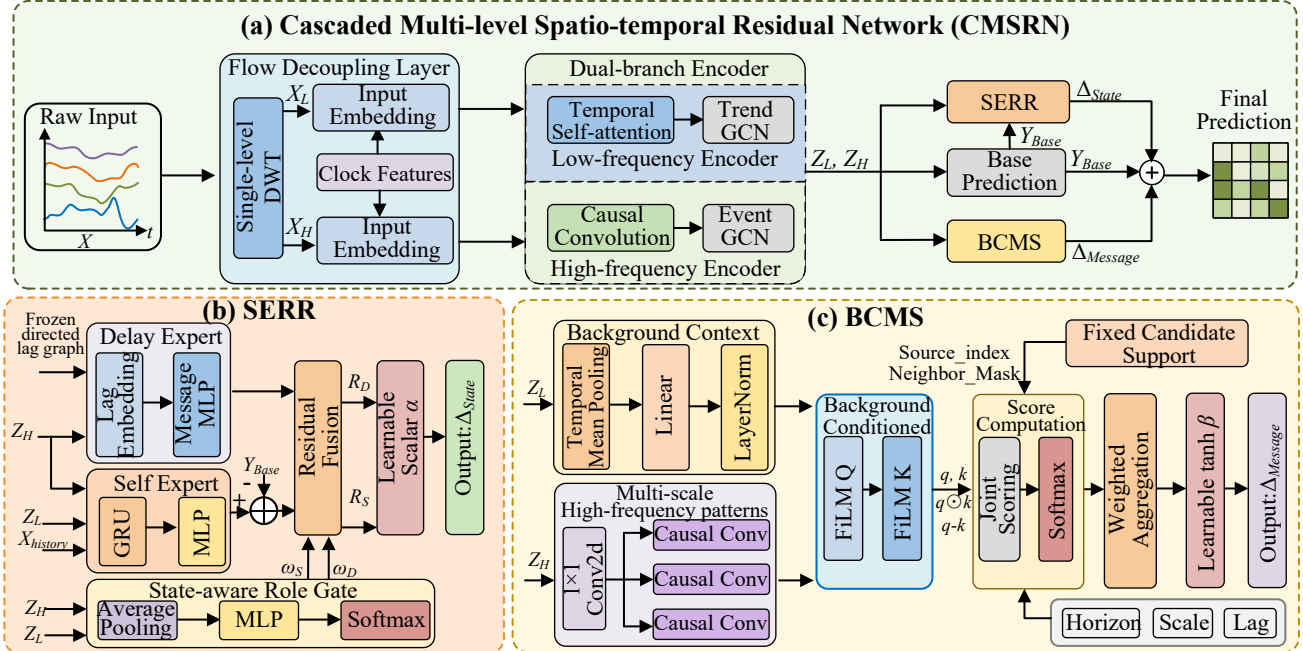

(a) Cascaded Multi-level Spatio-temporal Residual Network (CMSRN)
Raw Input
X
t
Flow Decoupling Layer
Single-level DWT
X_L
X_H
Input Embedding
Clock Features
Input Embedding
Dual-branch Encoder
Temporal Self-attention
Trend GCN
Low-frequency Encoder
Causal Convolution
Event GCN
High-frequency Encoder
Z_L, Z_H
SERR
Δ_State
Y_Base
Base Prediction
Y_Base
BCMS
Δ_Message
Final Prediction
(b) SERR
Frozen directed lag graph
Delay Expert
Lag Embedding
Message MLP
Z_H
Self Expert
GRU
MLP
Z_L
X_history
Y_Base
Residual Fusion
R_D
R_S
Learnable Scalar α
Output: Δ_State
ω_S
ω_D
State-aware Role Gate
Z_H
Z_L
Average Pooling
MLP
Softmax
(c) BCMS
Background Context
Z_L
Temporal Mean Pooling
Linear
LayerNorm
Multi-scale High-frequency patterns
Z_H
1×1 Conv2d
Causal Conv
Causal Conv
Causal Conv
Background Conditioned
FiLM Q
FiLM K
q, k
q⊙k
q-k
Source_index
Neighbor_Mask
Fixed Candidate Support
Score Computation
Joint Scoring
Softmax
Weighted Aggregation
Learnable tanh β
Output: Δ_Message
Horizon
Scale
Lag